

Interpreting the video feed

It is one thing to recognize an object and it is another to keep recognizing multiple objects multiple times every second when more can appear any time and current ones can disappear, move and rotate. Object recognition, although forms the base of the recognition based AR, it is still half the story told. The other half lies in interpreting the video feed.

Videos are constructed by taking pictures at a very fast rate. For a video to be watchable, it should exceed the rate of 25 frames per second. Translate this requirement for AR application and it means that the application should not take more than 1/25th of a second to identify objects. The creation of augmentation layer must also take place within the same time frame.

The constraints call in for innovation of better hardware as well as more efficient algorithms at all levels. Ideally an algorithm using dedicated graphics hardware can do pattern recognition at a much faster rate than a completely CPU based algorithm. One method used for recognition based AR is to identify the object once and track its movement as it happens. The comparison of previous image frame with current one can suggest displacement and rotation of objects. In many cases a 3D wireframe model is present within the application to make the comparison which helps understand the differences better.

Building 3D models

By now, it would be obvious that a large part of recognition based AR implementation depends on the 3D models of objects. Building a sufficiently large database of object data is required to provide a better AR experience. While images are easier to insert (all you need to do is to take pictures), 3D models can be difficult. One needs to design the model of the object before it could be fed into the database. Sure, there are tools available for the job. But the variation of design for even simple objects would lead to an enormous number of models. Modelling all of that into 3D is a daunting task. However with tools such as VideoTrace (<http://punchcard.com.au/>), the amount of effort reduces.

Together all these technologies and methods help in recognising the object that lies in front of the camera. Recognising the object is the core need of recognition based AR. The second part - overlaying the object with another object or presenting extra information about the object is easier. It involves creating a transparent layer on top of the actual view and then filling it with required information.